

LUND
UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written Examination
Number Theory
Thursday January 15, 2015
Duration: 8.00–13.00

Solutions may of course be presented in Swedish! The sheets of the department should be used. Write your initials and at most one solution on each sheet handed in. Give careful motivations to your solutions!

1. Find all solutions to the system of simultaneous congruences

$$\begin{cases} x \equiv 1 \pmod 7 \\ x \equiv 3 \pmod 8 \\ x \equiv 5 \pmod 9. \end{cases}$$

2. If p and q are distinct primes, prove that for any integer a,

$$pq \mid a^{pq} - a^p - a^q + a.$$

3. Let p be an odd prime, e a positive integer, and a an integer such that $\gcd(a, p) = 1$. Show that

$$x^2 \equiv a \pmod{p^e}$$

has either no solution or exactly two incongruent solutions $\pmod{p^e}$.

4. Let n be a positive integer. Show that the number of ordered pairs of positive integers with least common multiple n is $\tau(n^2)$ (which denotes the number of positive divisors of n^2).

5. Let p be an odd prime such that

$$p \mid a^2 + b^2$$

where $\gcd(a, b) = 1$. Prove that $p \equiv 1 \pmod 4$.

6. Let α be an irrational real number, and let $(p_k/q_k)_{k=0}^{\infty}$ denote the convergents to the simple continued fraction of α. Show that for $k \geq 0$ either

$$\left| \alpha - \frac{p_k}{q_k} \right| < \frac{1}{2q_k^2}$$

or

$$\left| \alpha - \frac{p_{k+1}}{q_{k+1}} \right| < \frac{1}{2q_{k+1}^2}.$$